

Identification using instrumental variables

Maximilian Kasy

Department of Economics, University of Oxford

Fall 2025

Takeaways for this part of class

1. The modern approach to identification allows for effect heterogeneity.
2. With heterogeneity, 2SLS estimates a weighted average treatment effect, the LATE.
3. If we insist on identifying the average treatment effect (ATE), then instrumental variables only allow us to identify bounds (“partial identification”).

Outline

- The canonical framework with effect heterogeneity and binary treatment and instrument.
- Identification of the LATE.
- Partial identification of the ATE.

Instrumental variables

- recall: simultaneous equations models with exclusion restrictions
- $\Rightarrow$ instrumental variables

$$\beta = \frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, D)}.$$

- we will now give a new interpretation to β
- using the potential outcomes framework, allowing for heterogeneity of treatment effects
- “Local Average Treatment Effect” (LATE)

6 assumptions

Angrist, J., Imbens, G., and Rubin, D. (1996). Identification of causal effects using instrumental variables. *Journal of the American Statistical Association*, 91(434):444–455.

1. $Z \in \{0, 1\}, D \in \{0, 1\}$
2. $Y = D \cdot Y^1 + (1 - D) \cdot Y^0$
3. $D = Z \cdot D^1 + (1 - Z) \cdot D^0$
4. $D^1 \geq D^0$
5. $Z \perp (Y^0, Y^1, D^0, D^1)$
6. $\text{Cov}(Z, D) \neq 0$

Discussion of assumptions

Generalization of randomized experiment

- D is “partially randomized”
- instrument Z is randomized
- D depends on Z , but is not fully determined by it

1. Binary treatment and instrument:

both D and Z can only take two values

results generalize, but things get messier without this

2. Potential outcome equation for Y : $Y = D \cdot Y^1 + (1 - D) \cdot Y^0$

- *exclusion restriction*: Z does not show up in the equation determining the outcome.
- “*stable unit treatment values assumption*” (SUTVA): outcomes are not affected by the treatment received by other units.
excludes general equilibrium effects or externalities.

3. **Potential outcome equation for D :** $D = Z \cdot D^1 + (1 - Z) \cdot D^0$

SUTVA; treatment is not affected by the instrument values of other units

4. **No defiers:** $D^1 \geq D^0$

- four possible combinations for the potential treatments (D^0, D^1) in the binary setting
- $D^1 = 0, D^0 = 1$, is excluded
- $\Leftrightarrow$ monotonicity

Table: No defiers

	D^0	D^1
Never takers (NT)	0	0
Compliers (C)	0	1
Always takers (AT)	1	1
Defiers	1	0

5. **Randomization:** $Z \perp (Y^0, Y^1, D^0, D^1)$

- Z is (as if) randomized.
- in applications, have to justify both exclusion and randomization
- no reverse causality, common cause!

6. **Instrument relevance:** $\text{Cov}(Z, D) \neq 0$

- guarantees that the IV estimand is well defined
- there are at least some compliers
- testable
- near-violation: weak instruments

Graphical illustration

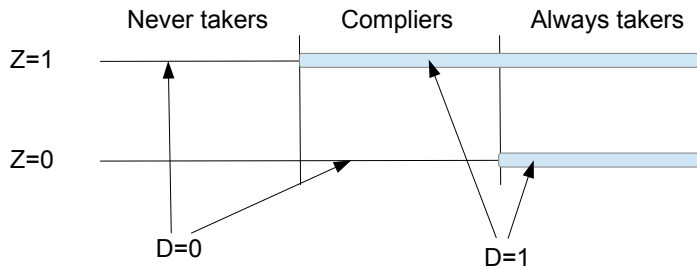


Illustration explained

- 3 groups, never takers, compliers, and always takers
- by randomization of Z :
each group represented equally given $Z = 0$ / $Z = 1$
- depending on group:
observe different treatment values and potential outcomes.
- will now take the IV estimand
$$\frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, D)}$$
- interpret it in terms of potential outcomes:
average causal effects for the subgroup of compliers
- idea of proof:
take the “top part” of figure 14, and subtract the “bottom part.”

Preliminary result:

If Z is binary, then

$$\frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, D)} = \frac{E[Y|Z=1] - E[Y|Z=0]}{E[D|Z=1] - E[D|Z=0]}.$$

Practice problem

Try to prove this!

Proof

- Consider the covariance in the numerator:

$$\begin{aligned}\text{Cov}(Z, Y) &= E[YZ] - E[Y] \cdot E[Z] \\ &= E[Y|Z = 1] \cdot E[Z] - (E[Y|Z = 1] \cdot E[Z] - E[Y|Z = 0] \cdot E[1 - Z]) \cdot E[Z] \\ &= (E[Y|Z = 1] - E[Y|Z = 0]) \cdot E[Z] \cdot E[1 - Z].\end{aligned}$$

- Similarly for the denominator:

$$\text{Cov}(Z, D) = (E[D|Z = 1] - E[D|Z = 0]) \cdot E[Z] \cdot E[1 - Z].$$

- The $E[Z] \cdot E[1 - Z]$ terms cancel when taking a ratio

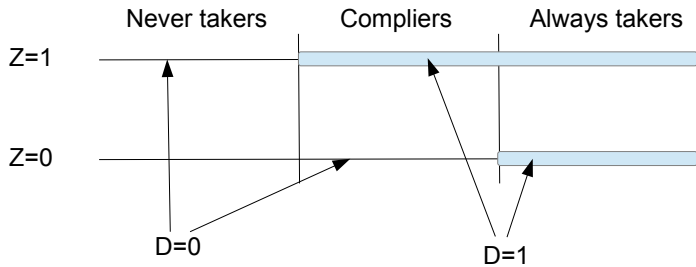
The “LATE” result

$$\frac{E[Y|Z=1] - E[Y|Z=0]}{E[D|Z=1] - E[D|Z=0]} = E[Y^1 - Y^0 | D^1 > D^0]$$

Practice problem

Try to prove this!

Hint: decompose $E[Y|Z=1] - E[Y|Z=0]$ in 3 parts corresponding to our illustration



Proof

- “top part” of figure:

$$\begin{aligned} E[Y|Z = 1] &= E[Y|Z = 1, NT] \cdot P(NT|Z = 1) \\ &\quad + E[Y|Z = 1, C] \cdot P(C|Z = 1) \\ &\quad + E[Y|Z = 1, AT] \cdot P(AT|Z = 1) \\ &= E[Y^0|NT] \cdot P(NT) + E[Y^1|C] \cdot P(C) + E[Y^1|AT] \cdot P(AT). \end{aligned}$$

- first equation relies on the no defiers assumption
- second equation uses the exclusion restriction and randomization assumptions.
- Similarly

$$\begin{aligned} E[Y|Z = 0] &= E[Y^0|NT] \cdot P(NT) + \\ &\quad E[Y^0|C] \cdot P(C) + E[Y^1|AT] \cdot P(AT). \end{aligned}$$

proof continued:

- Taking the difference, only the complier terms remain, the others drop out:

$$E[Y|Z = 1] - E[Y|Z = 0] = (E[Y^1|C] - E[Y^0|C]) \cdot P(C).$$

- denominator:

$$\begin{aligned} E[D|Z = 1] - E[D|Z = 0] &= E[D^1] - E[D^0] \\ &= (P(C) + P(AT)) - P(AT) = P(C). \end{aligned}$$

- taking the ratio, the claim follows. $\square$

Recap

LATE result:

- take the **same statistical object** economists estimate a lot
- which used to be interpreted as average treatment effect
- **new interpretation** in a more general framework
- allowing for heterogeneity of treatment effects
- $\Rightarrow$ treatment effect for a subgroup

Practice problem

Is the LATE, $E[Y^1 - Y^0 | D^1 > D^0]$, a structural object?

Empirical example

- **Angrist, J.D. and Krueger, A.B. (1991).** Does compulsory school attendance affect schooling and earnings? *The Quarterly Journal of Economics*, 106(4):979–1014.
- compare individuals born in different quarters of the year
- school start age and structure compulsory schooling laws
- $\Rightarrow$ people born late in the year have to stay in school longer
- quarter of birth as an instrument for educational attainment in estimates of returns to schooling
- estimates effect for those affected by compulsory schooling laws

Practice problem

Who do you think are the compliers for the quarter of birth instrument?

TABLE III
 PANEL A: WALD ESTIMATES FOR 1970 CENSUS—MEN BORN 1920–1929^a

	(1) Born in 1st quarter of year	(2) Born in 2nd, 3rd, or 4th quarter of year	(3) Difference (std. error) (1) – (2)
ln (wkly. wage)	5.1484	5.1574	–0.00898 (0.00301)
Education	11.3996	11.5252	–0.1256 (0.0155)
Wald est. of return to education			0.0715 (0.0219)
OLS return to education ^b			0.0801 (0.0004)

Panel B: Wald Estimates for 1980 Census—Men Born 1930–1939

	(1) Born in 1st quarter of year	(2) Born in 2nd, 3rd, or 4th quarter of year	(3) Difference (std. error) (1) – (2)
ln (wkly. wage)	5.8916	5.9027	–0.01110 (0.00274)
Education	12.6881	12.7969	–0.1088 (0.0132)
Wald est. of return to education			0.1020 (0.0239)
OLS return to education			0.0709 (0.0003)

An alternative approach: Bounds

- keep the **old structural object** of interest: average treatment effect
- but analyze its identification in the more general framework with heterogeneous treatment effects
- in general: we can learn something, not everything
- $\Rightarrow$ bounds / “**partial identification**”

Manski, C. (2003). *Partial identification of probability distributions*. Springer Verlag.

Same assumptions as before

1. $Z \in \{0, 1\}, D \in \{0, 1\}$
2. $Y = D \cdot Y^1 + (1 - D) \cdot Y^0$
3. $D = Z \cdot D^1 + (1 - Z) \cdot D^0$
4. $D^1 \geq D^0$
5. $Z \perp (Y^0, Y^1, D^0, D^1)$
6. $\text{Cov}(Z, D) \neq 0$

additionally:

7. Y is bounded, $Y \in [0, 1]$

Decomposing ATE in known and unknown components

- decompose $E[Y^1]$:

$$E[Y^1] = E[Y^1|NT] \cdot P(NT) + E[Y^1|C \vee AT] \cdot P(C \vee AT).$$

- terms that are identified:

$$E[Y^1|C \vee AT] = E[Y|Z = 1, D = 1]$$

$$P(C \vee AT) = E[D|Z = 1]$$

$$P(NT) = E[1 - D|Z = 1]$$

and thus

$$E[Y^1|C \vee AT] \cdot P(C \vee AT) = E[YD|Z = 1].$$

- Data tell us nothing about $E[Y^1|NT]$.
 Y^1 is never observed for never takers.
- but we know, since Y is bounded, that

$$E[Y^1|NT] \in [0, 1]$$

- Combining these pieces, get upper and lower bounds on $E[Y^1]$:

$$E[Y^1] \in [E[YD|Z = 1], \\ E[YD|Z = 1] + E[1 - D|Z = 1]].$$

- For Y^0 , similarly

$$E[Y^0] \in [E[Y(1-D)|Z=0], \\ E[Y(1-D)|Z=0] + E[D|Z=0]].$$

- Data are uninformative about $E[Y^0|AT]$.

Practice problem

Show this.

Combining to get bounds on ATE

- lower bound for $E[Y^1]$, upper bound for $E[Y^0] \Rightarrow$ lower bound on $E[Y^1 - Y^0]$

$$E[Y^1 - Y^0] \geq E[YD|Z = 1] - E[Y(1 - D)|Z = 0] - E[D|Z = 0]$$

- upper bound for $E[Y^1]$, lower bound for $E[Y^0]$
 $\Rightarrow$ upper bound on $E[Y^1 - Y^0]$

$$E[Y^1 - Y^0] \leq E[YD|Z = 1] - E[Y(1 - D)|Z = 0] + E[1 - D|Z = 1]$$

Between randomized experiments and nothing

- bounds on ATE:

$$E[Y^1 - Y^0] \in [E[YD|Z = 1] - E[Y(1 - D)|Z = 0] - E[D|Z = 0], \\ E[YD|Z = 1] - E[Y(1 - D)|Z = 0] + E[1 - D|Z = 1]].$$

- length of this interval:

$$E[1 - D|Z = 1] + E[D|Z = 0] = P(NT) + P(AT) = 1 - P(C)$$

- Share of compliers $\rightarrow 1$
 - interval (“identified set”) shrinks to a point
 - In the limit, $D = Z$
 - thus $(Y^1, Y^0) \perp D$ – randomized experiment
- Share of compliers $\rightarrow 0$
 - length of the interval goes to 1
 - In the limit the identified set is the same as without instrument

References

- Angrist, J., Imbens, G., and Rubin, D. (1996). *Identification of causal effects using instrumental variables*. Journal of the American Statistical Association, 91(434):444–455.
- Manski, C. F. (2003). Partial identification of probability distributions. Springer Verlag.